Expected Visiting Times of Interval Markov Chains

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Abstract—test

1 Introduction

Thema einführen

2 Preliminaries

2.1 Notation

We denote the entry in the i-th row and j-th column of a matrix $\mathcal{T}$, as $(\mathcal{T})_{ij}$. We will use the notation $\underline{n} = 1, 2, \ldots, n$ with $n \in \mathbb{N}$ for sets. Cardinality of set S as $|S|$.

◯ ◊

2.2 Matheteil

Linear Order Measurable set (used in **??**) Iverson Brackets

Definition 2.1 (Row-Stochastic Matrix)**.** *A square matrix $\mathcal{T} \in \mathbb{R}^{n \times n}$ is row-stochastic, if and only if $\sum_{j=0}^{n} (\mathcal{T})_{ij} = 1$ for all rows $i \in \underline{n}$*

When considering each row as a function $f_i(j) = (\mathcal{T})_{ij}$, these functions are discrete probability distributions. Expected Values

Definition 2.2 (Discrete Probability Distribution)**.** *A function P is a discrete probability distribution, if $P : X \to [0, 1]$ where X is a countable set and $\sum_{x \in X} P(x) = 1$.*

2.3 Markov Chains

OMISSION OF $\mathcal{M}$

In this Section, we provide definitions and ... for Markov Chains, as well as Interval Markov Chains. The notation is based on **[?]**, **[?]**...

2.3.1 Discrete-Time (Interval) Markov Chain

Definition 2.3 (DTMC)**.** *A discrete-time Markov chain (DTMC) is a tuple $\mathcal{M} = (S, \mathbf{P}, \iota_{init})$, where*

(i) S is a countable, nonempty set of states,
(ii) $\mathbf{P} : S \times S \to [0, 1]$ is the transition probability function, such that $\sum_{s' \in S} \mathbf{P}(s, s') = 1$ for all states $s \in S$, and
(iii) $\iota_{init} : S \to [0, 1]$ is the initial distribution, such that $\sum_{s \in S} \iota_{init}(s) = 1$.

We assume in this thesis, that the DTMCs are finite, i.e., that the set of states S is finite. Furthermore, we assume an arbitrary but fixed linear order $\prec$ on the set S, such that the states $s \in S$ can be indexed. Thus, $s_i \prec s_j$ if $s, s' \in S$ and $i < j$. Due to the conditions (i) and (ii) provided in **??**, the transition probability function $\mathbf{P}$ and the initial distribution ι_{init} are discrete probability distributions.

??(notion of a current state) Moreover, if $\mathbf{P}(s, s') > 0$ for two given $s, s' \in S$, then it is possible to move from s to s' in one step. We refer to these states s' as direct successors of s and to s as a direct predecessor of all states s'. Additionally, we define a neighbour set N, with $s' \in N(s)$, if and only if

PROBABILITY SPACE OF DTMC

Definition 2.4 (Transition Matrix)**.** *The transition matrix $\mathcal{T}^{\mathcal{M}}$ of a DTMC $\mathcal{M}$ is a square matrix with size $S \times S$, such that $(\mathcal{T}^{\mathcal{M}})_{ij} = \mathbf{P}(s_i, s_j)$.*

gemeint S klar?

Notice that due to the condition, given in (ii) of **??**, that $\sum_{s_j \in S} \mathbf{P}(s_i, s_j) = 1$ for all states $s \in S$, by defintion $\sum_{j=0}^{|S|} (\mathcal{T})_{ij} = 1$ for all rows $i \in \underline{n}$. Thus, the transition matrix $\mathcal{T}$ of a DTMC $\mathcal{M}$ is row-stochastic, see **??**.

Definition 2.5. *Every DTMC $\mathcal{M}$ induces a directed weigthed graph $\mathcal{G}^{\mathcal{M}} = (V, E, f)$, with the nodes $V = S^{\mathcal{M}}$, the edges $(s, s') \in E$ if and only if $\mathbf{P}(s, s') > 0$, and the weight function $f(s, s') = \mathbf{P}(s, s')$*

simply use **P** instead of f?

Definition 2.6 (Path)**.** *An infinite path π in a DTMC $\mathcal{M}$ is a sequence $\pi = s_i \to s_j \to \ldots$ of states with $s_i, s_j \in S$, such that $\mathbf{P}(\pi[k], \pi[k+1]) > 0$, where $\pi[k] \in S$ denotes the k-th element of the sequence/path for all $k \in \mathbb{N}_0$. A finite path is a finite prefix of an infinite path.*

Prob distr. of paths

Definition 2.7 ((Step-Bounded) Constrained Reachability (Probabilities?))**.** *Let $T, R \subseteq S^{\mathcal{M}}$ be two sets of states in a DTMC $\mathcal{M}$. We now define the probability of only visiting states in R until eventually a state in T is visited as $Pr(R \cup T)$, where*

$$R \cup T = \{\pi \in Paths^{\mathcal{M}}_{inf} | \exists k \in \mathbb{N} : \pi[k] \in T \wedge \forall i < k : \pi[i] \in R\}$$

Furthermore, the probability of only visint states in R until a state in T is visited at step $k \in \mathbb{N}$ is given by $Pr(R \cup^{=k}) \ldots$ (step-bounded defs necessary?)

Definition 2.8 (Reachability Probability)**.** *The reachability probability of a set $T \subseteq S^{\mathcal{M}}$ in a DTMC $\mathcal{M}$ is given by $Pr(\Diamond T)$, where $\Diamond T = S \cup T$*

As $S^{\mathcal{M}}$ defines all states in $\mathcal{M}$, this simply defines the probability of visiting a state $t \in T$ at an arbitrary step in

the future, i.e. if $Pr(\Diamond T) > 0$ at least one state $t \in T$ is reachable. In case $T = \{t\}$ is a singleton set, we omit the braces and denote $\Diamond t$ instead of $\Diamond\{t\}$ (and $S \cup t$ instead of $S \cup T$)

$\sum_{Paths_{inf}^{\mathcal{M}}}$ DEFINIEREN (?)

We use the LTL unary prefix operator $\bigcirc$ to denote the temporal modality "next", meaning

Definition 2.9 (Next Step Probability). *Let $\mathcal{M}$ be a DTMC and let $E \in Paths_{inf}^{\mathcal{M}}$ be a (countable) set of infinite paths in $\mathcal{M}$. We now define:*

$$\bigcirc E = \{\pi[2] \to \cdots \in Paths_{inf}^{\mathcal{M}} | \pi[1] \to \pi[2] \to \cdots \in E\}$$

The next step probability with respect to E is then defined as

$$Pr(\bigcirc E) = Pr()$$

Definition 2.10 ((Bottom) Strongly Connected Component). *A set of states $C \subseteq S^{\mathcal{M}}$ of a DTMC $\mathcal{M}$ is strongly connected if and only if for every pair of states $s, t \in S$, there exists a path $\pi \in Paths_{inf}^{\mathcal{M}}$, such that $\pi[0] = s$ and $\pi[k]$ for some $k \in \mathbb{N}$.*
If C is strongly connected and no proper subset, i.e. $C' \subset C$, is strongly connected, then C is called a strongly connected component (SCC).
Furthermore, an SCC C is called a bottom stronlgy connected component (BSCC) if there exists no $k \in \mathbb{N}$ and no path $\pi[0] \to \cdots \to \pi[k] \in Paths_{inf}^{\mathcal{M}}$ such that $\pi[0] \in C$ and $\pi[k] \notin C$.

In other words, once the model enters a BSCC, it cannot leave it again.

SCC (minimal amount of states of set such that still strongly connec) (Acyclic DTMC)

The states can be classified ...

Definition 2.11 (Transient and recurrent states). *A state $s \in S$ is called recurrent if and only if $Pr_s(\bigcirc\Diamond s) = 1$. All states that are not recurrent are called transient.*

In other words, a state is recurrent precisely when we are guaranteed to reach it from all of its direct successors, meaning once we reach a recurrent state, we are bound to return to it. (Thus reaching it infinitely often) Using these definitions, we can split the states S of a DTMC into two sets with S_{tr} containing all of the transient states and S_{re} containing all of the recurrent states.

Definition 2.12 (Absorbing state). *A states s of a DTMC $\mathcal{M}$ is refered to as an absorbing state if and only if $\mathbf{P}^{\mathcal{M}}(s, s) = 1$.*

This means, once the model enters an absorbing state, it cannot leave it. Notice, that absorbing states are a special type of recurrent states, as the only direct successor of an absorbing state $s \in S$ is the s itself which trivially has $Pr_s(\Diamond s) = 1$.

Using BSCCs we can define a special class of DTMCs.

Definition 2.13 (Absorbing DTMC). *A DTMC $\mathcal{M}$ is called absorbing if and only if all BSSCs are singleton sets consisting of an absorbing state.*

Definition 2.14 (Reducible DTMC). *A DTMC $\mathcal{M}$ is called reducible if and only if*

Theorem 2.1. *For any DTMC $\mathcal{M}$, $s \in S$ with $\iota_{init}(s) > 0$ it holds that $Pr(\Diamond S_{re}) = 1$*

(Limit Behaviour?)
Canonical Form
(Reward Structure)

2.3.2 Expected Visiting Times

Intuitively, the notion of EVTs describes how often we expect the model to be in a given state. To formally introduce EVTs, we fix a DTMC $\mathcal{M}$ and start by introducing *visiting times* along paths.

Definition 2.15 (Visiting Times of Infinite Paths). *The function $vt_s : Paths_{inf}^{\mathcal{M}} \to \mathbb{R}$ describes the visiting time of a state $s \in S$ along infinite paths in $\mathcal{M}$, where for $\pi \in Paths_{inf}^{\mathcal{M}}$*

$$vt_s(\pi) = \sum_{k=0}^{\infty} [\![\pi[k] = s]\!]$$

Borel Measurability

Definition 2.16 (Expected Visiting Times in DTMCs). *For a state $s \in S$, the expected visiting time of s is given by*

$$\mathbb{E}^{\mathcal{M}}[vt_s] = \int_{\pi \in Paths_{inf}^{\mathcal{M}}} vt_s(\pi) dPr^{\mathcal{M}}(\pi)$$

OMSSION OF $^{\mathcal{M}}$

Using the aforementioned classification of the states of $\mathcal{M}$ into transient and recurrent states in **??**, we can already give an expression for the latter.

Lemma 2.1 (EVT of Recurrent State). *For all recurrent states $s \in S_{re}$ in $\mathcal{M}$, it holds that*

$$\mathbb{E}[vt_s] = \begin{cases} \infty & \text{if } s \text{ is reachable,} \\ 0 & \text{else.} \end{cases}$$

(Proof??) Thus, we will focus on finding the non-trivial expected visiting times of the transient states of DTMCs.

EVTs Def. –¿ Wie berechnen

2.3.3 Interval Markov Chains

In this thesis, we will focus on Interval Markov Chains. These are defined as follows.

Definition 2.17 (Interval Markov Chain). *An Interval-valued Discrete-time Markov chain (IDTMC) is a tuple $\mathcal{I} = (S, \overline{\mathbf{P}}, \underline{\mathbf{P}}, \bar{\iota}_{init}, \underline{\iota}_{init})$, where*

(i) S is a countable, nonempty set of states,
(ii) $\overline{\mathbf{P}} : S \times S \to [0, 1]$ is the transition probability function, where $\overline{\mathbf{P}}(s, s')$ gives the upper bound on the transition probability from state s to s',
(iii) $\underline{\mathbf{P}} : S \times S \to [0, 1]$ is the transition probability function, where $\underline{\mathbf{P}}(s, s')$ gives the lower bound on the transition probability from state s to s',
(iv) $\bar{\iota}_{init} : S \to [0, 1]$ is the upper bound on the initial distribution, and
(v) $\underline{\iota}_{init} : S \to [0, 1]$ is the upper bound on the initial distribution.

IMC Def

There are two commonly used semantics, which we will consider in this paper. The first one is the *once-and-for-all*, see **[?]**, **[?]**, **[?]**.

EVTs für IMCs

Definition 2.18 (Once-and-for-all Semantic)**.** *An arbitrary DTMC $\mathcal{M}$ satisfies an IDTMC $\mathcal{I}$ if and only if $\underline{\mathbf{P}}^{\mathcal{I}}(s,s') \leq \mathbf{P}^{\mathcal{M}}(s,s') \leq \overline{\mathbf{P}}^{\mathcal{I}}(s,s')$ and $\underline{\iota}^{\mathcal{I}}_{init}(s,s') \leq \iota^{\mathcal{M}}_{init}(s,s') \leq \overline{\iota}^{\mathcal{I}}_{init}(s,s')$.*

Thus, any DTMC whose initial distribution and transition probabilities lie in the intervals defined by the IDTMC is satisfactory.

Definition 2.19 (IMDP semantic)**.** *An arbitrary DTMC $\mathcal{M} = (T, \mathbf{P}, \iota_{init})$ satisfies an IDTMC $\mathcal{I} = (S, \overline{\mathbf{P}}, \underline{\mathbf{P}}, \overline{\iota}_{init}, \underline{\iota}_{init})$ if and only if there exists an injective mapping $\tau : T \to S$ such that*

(i) $\tau(t) = s \in S$ *for all* $t \in T \implies \underline{\iota}_{init}(s) \leq \iota_{init}(t) \leq \overline{\iota}_{init}(s)$

(ii) $\tau(t) = s \in S$ *for all* $t \in T \implies \underline{\mathbf{P}}(s) \leq \mathbf{P}(t) \leq \overline{\mathbf{P}}(s)$

. . . examples intuition

(Explain how they describe/define it: the minor differences between them)

3 Conclusion and Outlook

References

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